Michaël **Ramamonjisoa**

AI Research Engineer at Meta AI (FAIR)

6 rue Ménars, 75002 Paris (France) | French

+ 33 6 61223866 | ramamonjisoa.michael@gmail.com | https://michaelramamonjisoa.github.io/ | MichaelRamamonjisoa

Education

IMAGINE team – Ecole des Ponts/LIGM
Paris area, France

PhD — *Oct. 2018 – Nov. 2022*

Disentangling for Scene Understanding from Images – **Supervisor: Prof. Vincent Lepetit**

- 3D Geometry Estimation, Scene Understanding, 3D Pose Estimation

Ecole Normale Supérieure Paris-Saclay
Paris area, France

Master's degree - Mathématiques Vision Apprentissage (MVA) — *2017 – 2018*

- **3D Computer Vision** (R. Marlet, P. Monasse, M. Aubry); **Audio signal Analysis, Indexing and Transformations** (R. Badeau, G. Richard); **Introduction to Digital Imaging** (J. Delon, Y. Gousseau); **Object Recognition and Computer Vision** (I. Laptev, J. Ponce, C. Schmid, J. Sivic); **Speech and Language Processing** (E. Dupoux, B. Sagot, N. Zeghidour);
- **High Dimension Statistical Learning** (S. Mallat); **Convex Optimization** (A. D'Aspremont); **Deep Learning** (V. Lepetit); **Kernel Methods for machine learning** (J. Mairal, J-P. Vert); **Probabilistic Graphical Models** (F. Bach, G. Obozinski); **Unsupervised Learning** (R. Vidal)

Projects:
Unsupervised Learning: Low Rank Matrix Completion, Face Clustering (Matlab)
Object Recognition and Computer Vision: Training CNNs using synthetic images of people (Torch, Python, Blender)
Introduction to Digital Imaging: Texture Synthesis using CNNs (Caffe, Python)

Imperial College London
London, United Kingdom

MSc Optics and Photonics — *2013 – 2014*

- Double Diploma in Engineering with IOGS
- **Optics**, **Photonics**, **Information theory**, **Biophotonics**

Institut d'Optique Graduate School Paristech (IOGS)
Paris area, France

Grande Ecole - 3rd year abroad — *2011 – 2014*

- Diploma in Engineering
- **Optics**, **Signal and image processing**, **Electronics**, **Physics**

Professional experience

Meta AI - FAIR (DINO team)
Paris, France

Senior AI Research Engineer — *Sep. – present*

Research in Computer Vision and Deep Learning

Meta – Reality Labs
Zurich, Switzerland

Research Engineering Intern — *Aug. – Dec*

Research in Computer Vision and Deep Learning

Niantic
London, United Kingdom

Research Intern — *Jul. – Nov. 2020*

Research in Computer Vision and Deep Learning – led to an accepted paper at CVPR 2021

Prophesee
Paris, France

Research Intern — *Apr. – Sept. 2018*

Double Frequency Tracking Using Event-Based Cameras

HGH Systèmes Infrarouges
Paris area, France

Project Engineer — *Jan. 2015 – Sept. 2017*

- Manager of the infrared testing software (Infratest) of HGH: designed signal and image processing algorithms for optronics systems

Thalès Research & Technology
Paris area, France

Intern — *June – Nov. 2014*

Optical Design of a Lidar; signal processing for imaging on Matlab

ONERA (The French Aerospace Lab)
Paris area, France

Intern — *May – July 2013*

Designed a 3D-model of an urban area using Cinema 4D, then tested ONERA's Radar imaging algorithms on the 3D model

Publications

- O. Simeoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. Yi, **M. Ramamonjisoa**, F. Massa, D. Haziza, L. Wehrstedt, J. Wang, T. Darcet, T. Moutakanni, L. Santana, C. Roberts, A. Vedaldi, J. Tolan, J. Brandt, C. Couprie, J. Mairal, H. Jegou, P. Labatut, P. Bojanowski, **DINOv3**, *TMLR*, 2026
- C. Jose, T. Moutakanni, D. Kang, F. Baldassarre, T. Darcet, H. Xu, D. Li, M. Szafraniec, **M. Ramamonjisoa**, M. Oquab, O. Siméoni, H. V. Vo, P. Labatut, P. Bojanowski, **DINOv2 Meets Text: A Unified Framework for Image- and Pixel-Level Vision-Language Alignment**, *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025
- M. Wysoczańska, O. Siméoni, **M. Ramamonjisoa**, A. Bursuc, P. Pérez, **CLIP-DINOiser: Teaching Clip a Few DINO Tricks**, *ECCV*, 2024
- M. Wysoczańska, **M. Ramamonjisoa**, T. Trzciński, O. Siméoni, **CLIP-DIY: Clip Dense Inference Yields Open-Vocabulary Semantic Segmentation for-free**, *Winter Conference on Applications of Computer Vision (WACV)*, 2024
- **M. Ramamonjisoa**, S. Stekovic, V. Lepetit, **MonteBoxFinder: Detecting and Filtering Primitives to Fit a Noisy Point Cloud**, *European Conference on Computer Vision*, 2022
- **M. Ramamonjisoa**, M. Firman, J. Watson, V. Lepetit, D. Turmukhambetov, **Single Image Depth Prediction with Wavelet Decomposition**, *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021
- **M. Ramamonjisoa***, Y. Du*, V. Lepetit, **Predicting Sharp and Accurate Occlusion Boundaries in Monocular Depth Estimation Using Displacement Fields**, *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020
- **M. Ramamonjisoa**, V. Lepetit, **SharpNet: Fast and Accurate Recovery of Occluding Contours in Monocular Depth Estimation**, *The IEEE International Conference on Computer Vision (ICCV) Workshops*, 2019
- G. Pitteri*, **M. Ramamonjisoa***, V. Lepetit, **On Object Symmetries and 6D Pose Estimation from Images**, *2019 International Conference on 3D Vision (3DV)*, 2019
- C. Barrat, T. Lepot, **M. Ramamonjisoa**, S. Fradcourt, **Extension to NIR and visible ranges of high-resolution relative spectral response measurement using Fourier Transform Infrared Spectrometer (FTIR) of CMOS FPAs**, *Proc. SPIE 10433*, Electro-Optical and Infrared Systems: Technology and Applications XIV, 1043316 (6 October 2017); doi: 10.1117/12.2278301
- C. Barrat, T. Lepot, **M. Ramamonjisoa**, S. Fradcourt, **A practical implementation of high resolution relative spectral response measurement of CMOS IRFPAs using Fourier Transform Infrared Spectrometer (FTIR)**, *Proc. SPIE 9987*, Electro-Optical and Infrared Systems: Technology and Applications XIII, 99870V (21 October 2016); doi: 10.1117/12.2242014

** denotes equal contribution*

Teaching

- **Computer Vision**, 1st year of Master practical sessions, Ecole des Ponts Paristech
- **Numerical Methods**, 1st year of Master practical sessions, Institut d'Optique Graduate School
- **Artificial Intelligence**, 1st year of Master practical sessions, Université de Bordeaux
- **Deep Learning**, 1st year of Master practical sessions, Bordeaux INP

Relevant skills

- **Computer skills**: Python, C++, Matlab, Pytorch, Keras, Tensorflow, Unix shell, Cinema 4D, Blender, SolidWorks, LaTeX
- **Languages**: French (native); English (fluent, TOEFL 108/120); Spanish (conversational); Malagasy (basic skills); Mandarin (basic skills)